All 29 rate variables | L=1, D=+1, S=1 | E1 -> E2 -> E3 -> E4

21 synchronously integrated rates 8 held encoding rates

R1: E1 R2: E2 R3: E3 R4: E4

Every rate variable (29 rows; no row rescaling)

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=1, D=+1, S=2 | E1 -> E2 -> E4 -> E5

21 synchronously integrated rates 8 held encoding rates

R1: E1 R2: E2 R3: E4 R4: E5

Every rate variable (29 rows; no row rescaling)

E1 E2 E3 E4 E5 E6 — Position E
D_CW D_CCW — Direction
S1 S2 — Step
J_CW1 J_CW2 J_CCW1 J_CCW2 — Joint
V_CW1 V_CW2 V_CCW1 V_CCW2 — Readout V
G_CW1 G_CW2 G_CCW1 G_CCW2 — Gate G
M1 M2 M3 M4 M5 M6 — Shared M
I_global — Global I

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=1, D=-1, S=1 | E1 -> E6 -> E5 -> E4

21 synchronously integrated rates 8 held encoding rates

R1: E1 R2: E6 R3: E5 R4: E4

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

Every rate variable (29 rows; no row rescaling)

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=1, D=-1, S=2 | E1 -> E6 -> E4 -> E3

21 synchronously integrated rates 8 held encoding rates

R1: E1 R2: E6 R3: E4 R4: E3

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

Every rate variable (29 rows; no row rescaling)

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=2, D=+1, S=1 | E2 -> E3 -> E4 -> E5

21 synchronously integrated rates 8 held encoding rates

R1: E2 R2: E3 R3: E4 R4: E5

Every rate variable (29 rows; no row rescaling)

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=2, D=+1, S=2 | E2 -> E3 -> E5 -> E6

21 synchronously integrated rates 8 held encoding rates

R1: E2 R2: E3 R3: E5 R4: E6

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

Every rate variable (29 rows; no row rescaling)

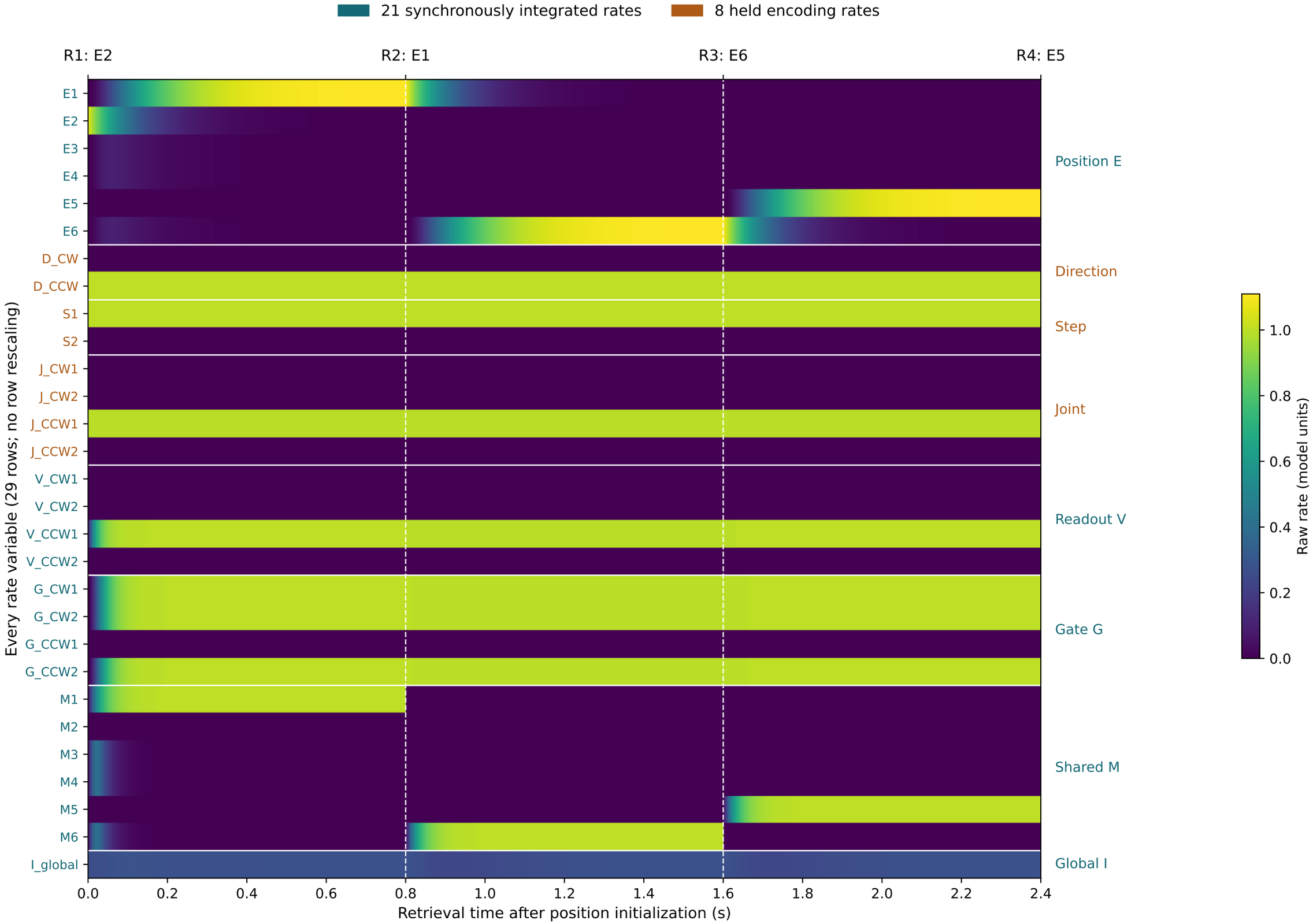
0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

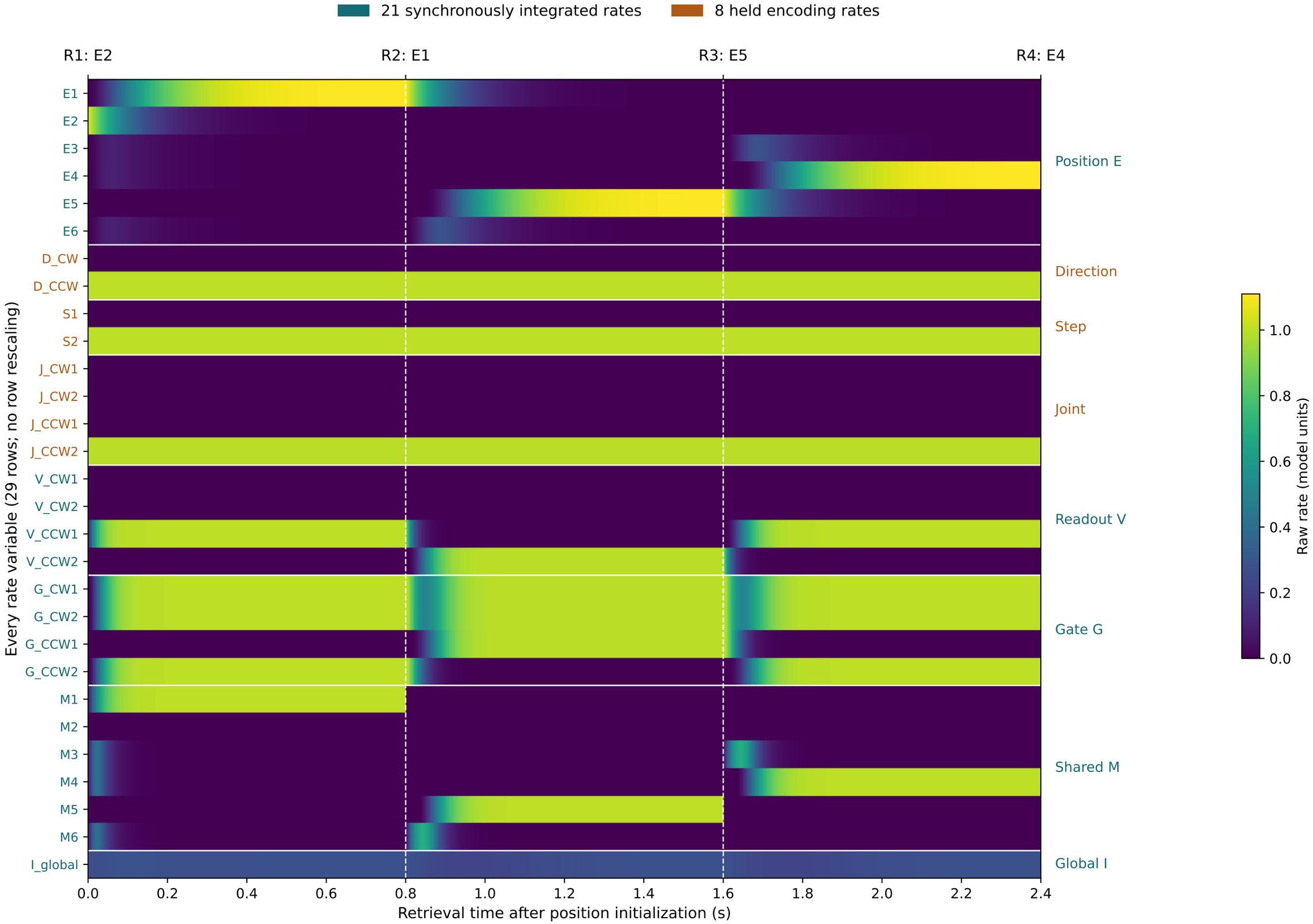
V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=2, D=-1, S=1 | E2 -> E1 -> E6 -> E5



V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=2, D=-1, S=2 | E2 -> E1 -> E5 -> E4



V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=3, D=+1, S=1 | E3 -> E4 -> E5 -> E6

21 synchronously integrated rates 8 held encoding rates

R1: E3 R2: E4 R3: E5 R4: E6

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

Every rate variable (29 rows; no row rescaling)

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=3, D=+1, S=2 | E3 -> E4 -> E6 -> E1

21 synchronously integrated rates 8 held encoding rates

R1: E3 R2: E4 R3: E6 R4: E1

Every rate variable (29 rows; no row rescaling)

E1 E2 E3 E4 E5 E6 — Position E
D_CW D_CCW — Direction
S1 S2 — Step
J_CW1 J_CW2 J_CCW1 J_CCW2 — Joint
V_CW1 V_CW2 V_CCW1 V_CCW2 — Readout V
G_CW1 G_CW2 G_CCW1 G_CCW2 — Gate G
M1 M2 M3 M4 M5 M6 — Shared M
I_global — Global I

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=3, D=-1, S=1 | E3 -> E2 -> E1 -> E6

21 synchronously integrated rates 8 held encoding rates

R1: E3 R2: E2 R3: E1 R4: E6

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

Every rate variable (29 rows; no row rescaling)

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=3, D=-1, S=2 | E3 -> E2 -> E6 -> E5

21 synchronously integrated rates 8 held encoding rates

R1: E3 R2: E2 R3: E6 R4: E5

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

Every rate variable (29 rows; no row rescaling)

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=4, D=+1, S=1 | E4 -> E5 -> E6 -> E1

21 synchronously integrated rates 8 held encoding rates

R1: E4 R2: E5 R3: E6 R4: E1

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

Every rate variable (29 rows; no row rescaling)

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=4, D=+1, S=2 | E4 -> E5 -> E1 -> E2

21 synchronously integrated rates 8 held encoding rates

R1: E4 R2: E5 R3: E1 R4: E2

Every rate variable (29 rows; no row rescaling)

E1 E2 E3 E4 E5 E6 — Position E
D_CW D_CCW — Direction
S1 S2 — Step
J_CW1 J_CW2 J_CCW1 J_CCW2 — Joint
V_CW1 V_CW2 V_CCW1 V_CCW2 — Readout V
G_CW1 G_CW2 G_CCW1 G_CCW2 — Gate G
M1 M2 M3 M4 M5 M6 — Shared M
I_global — Global I

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

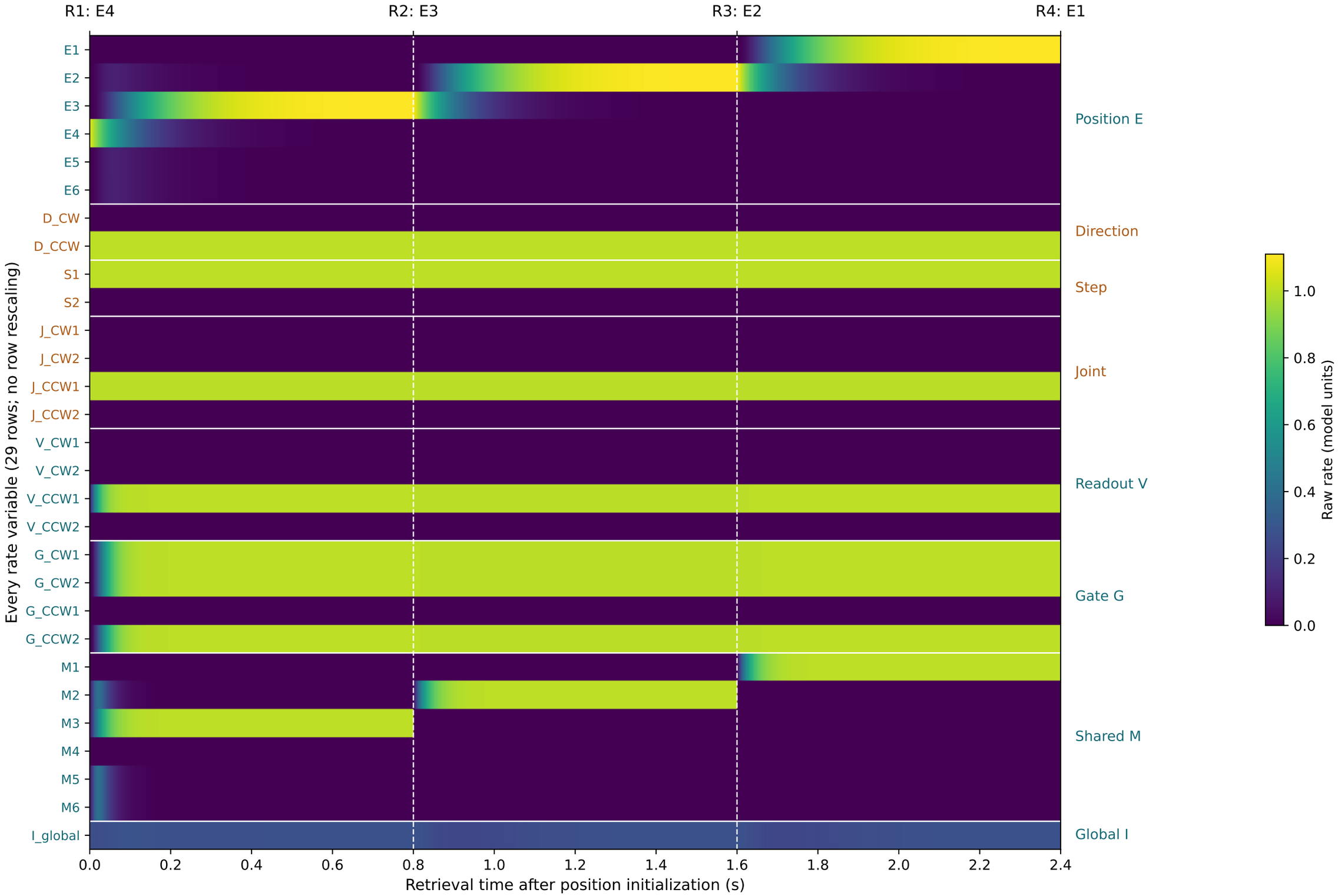
Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=4, D=-1, S=1 | E4 -> E3 -> E2 -> E1

21 synchronously integrated rates 8 held encoding rates



V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=4, D=-1, S=2 | E4 -> E3 -> E1 -> E6

21 synchronously integrated rates 8 held encoding rates

R1: E4 R2: E3 R3: E1 R4: E6

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

Every rate variable (29 rows; no row rescaling)

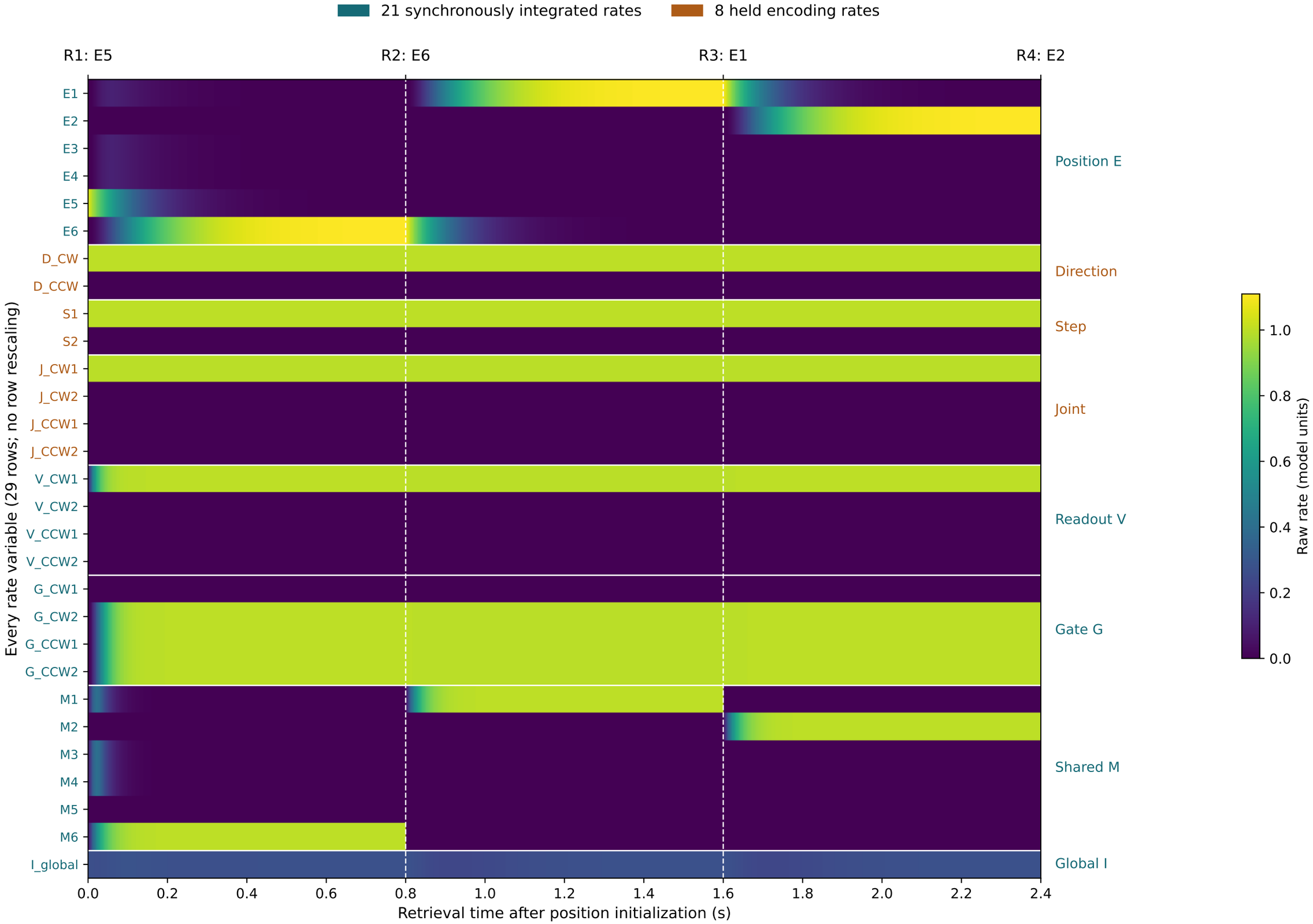
0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

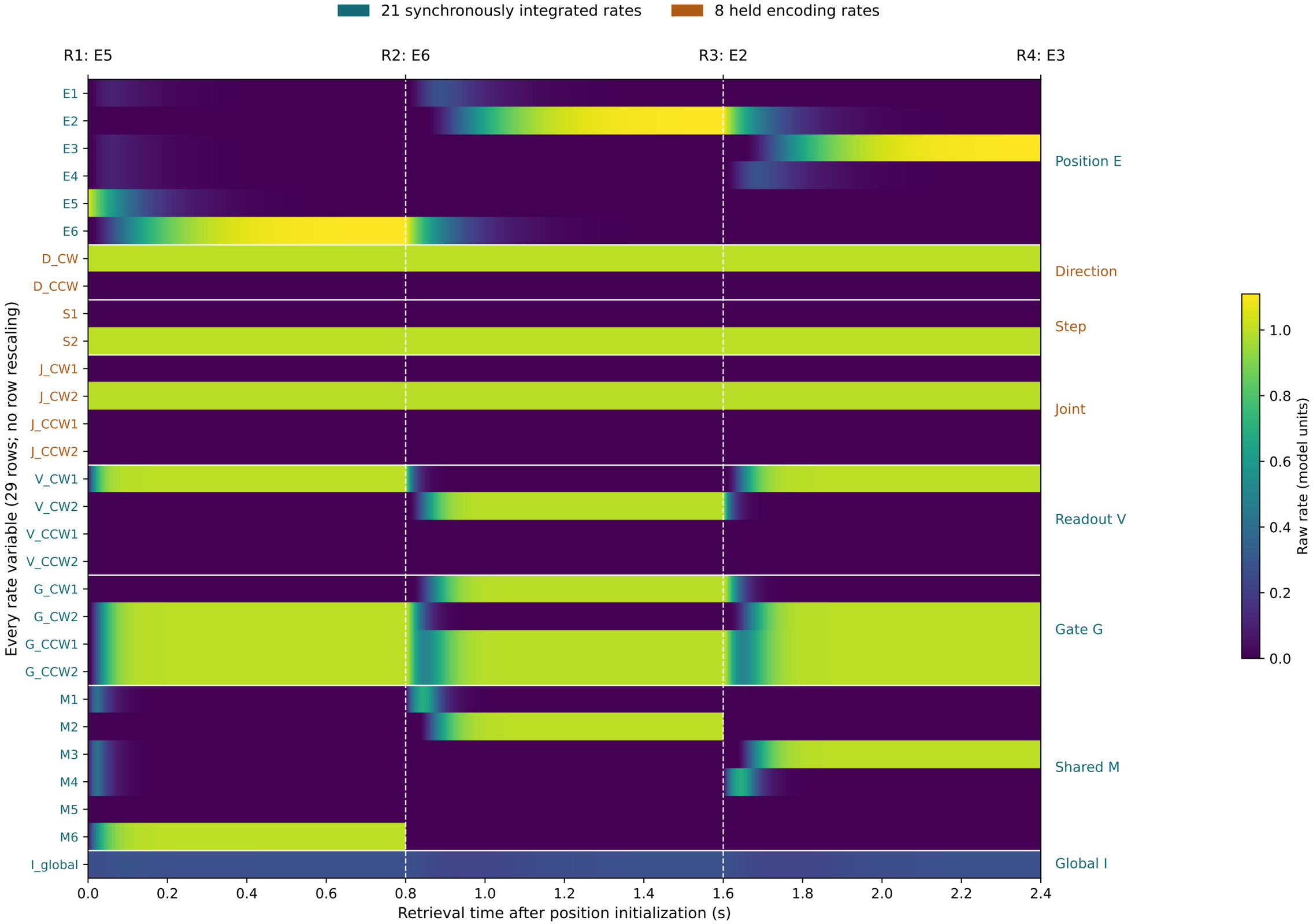
V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=5, D=+1, S=1 | E5 -> E6 -> E1 -> E2



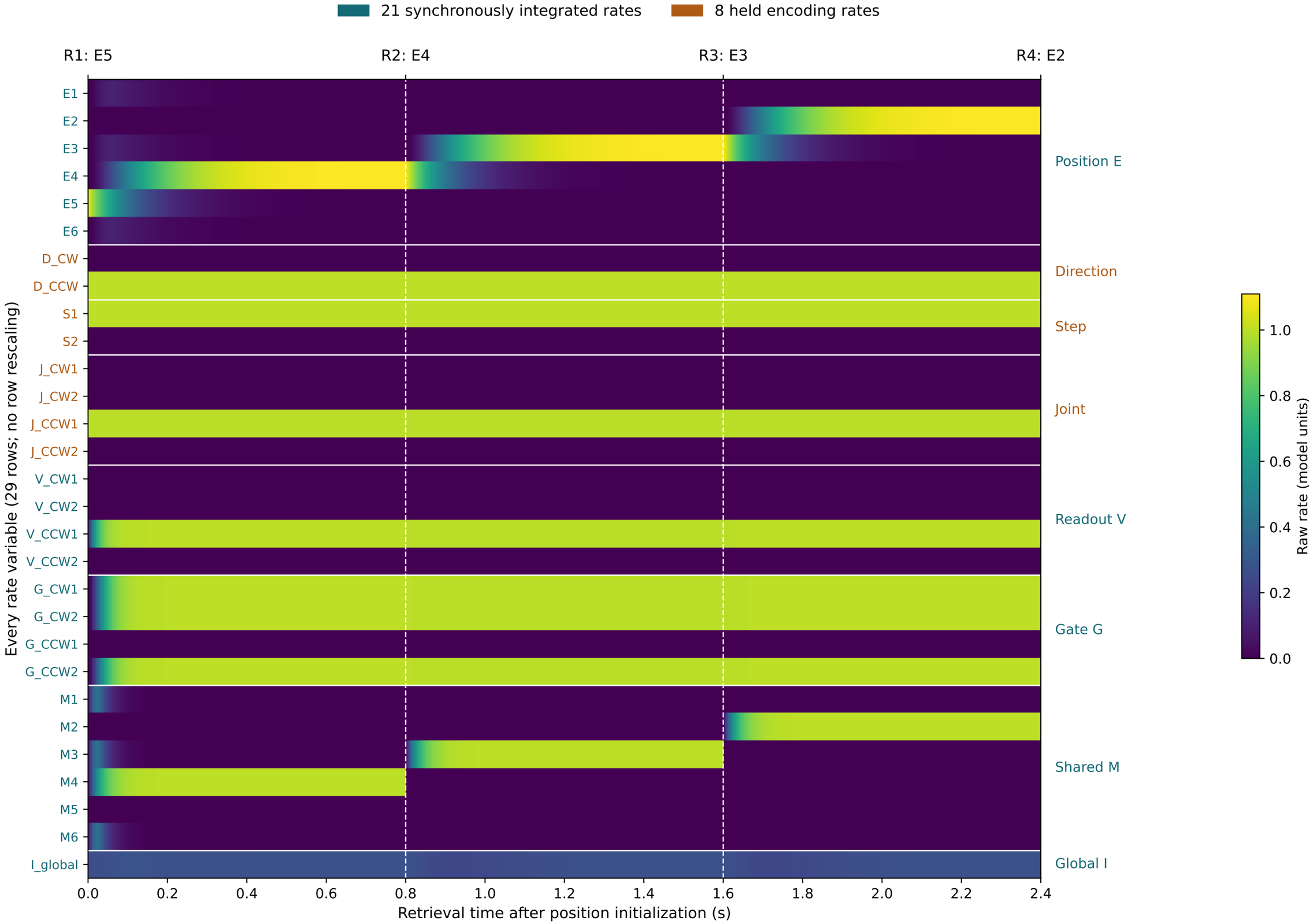
V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=5, D=+1, S=2 | E5 -> E6 -> E2 -> E3



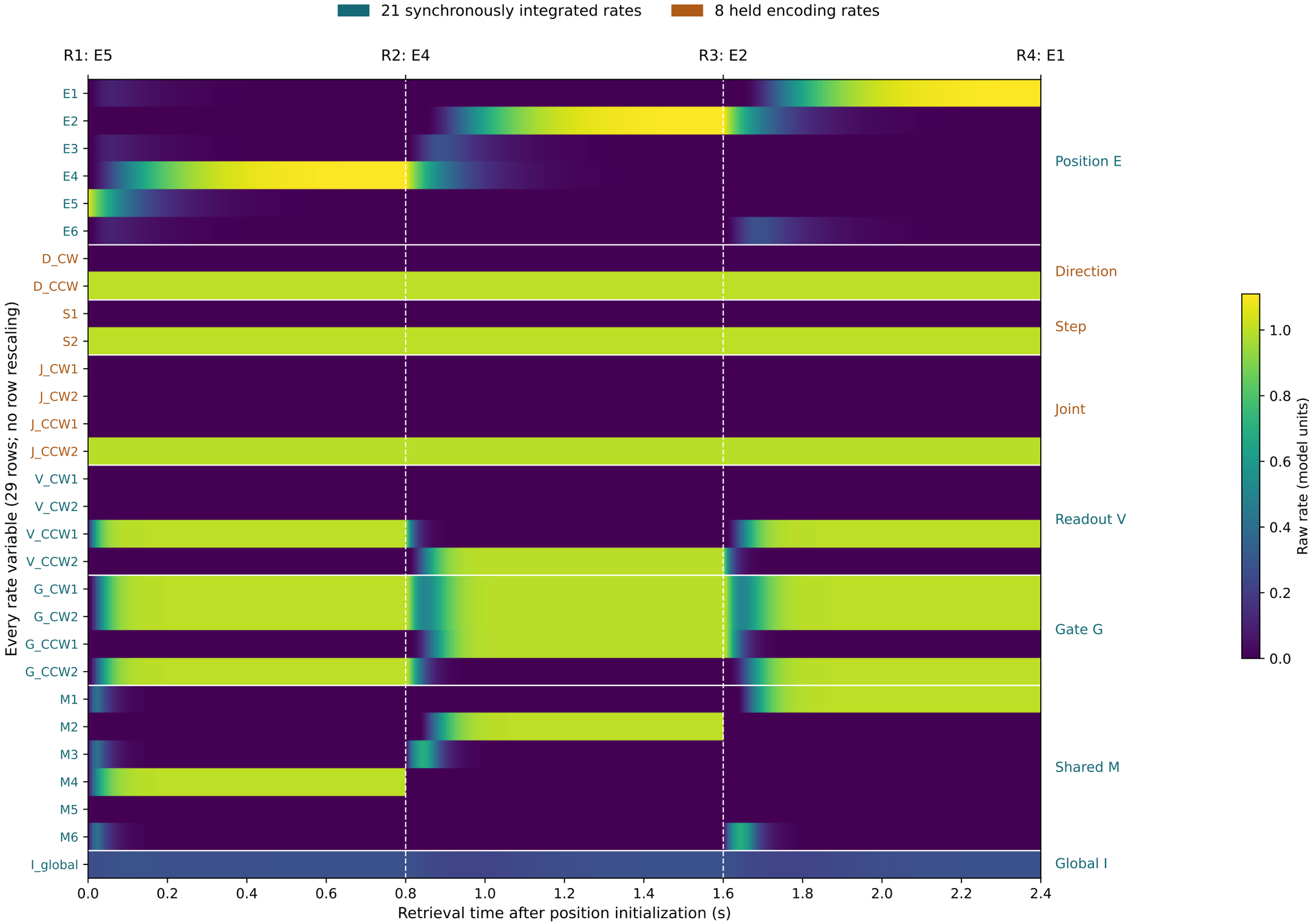
V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=5, D=-1, S=1 | E5 -> E4 -> E3 -> E2



V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=5, D=-1, S=2 | E5 -> E4 -> E2 -> E1



V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=6, D=+1, S=1 | E6 -> E1 -> E2 -> E3

21 synchronously integrated rates 8 held encoding rates

R1: E6 R2: E1 R3: E2 R4: E3

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

Every rate variable (29 rows; no row rescaling)

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=6, D=+1, S=2 | E6 -> E1 -> E3 -> E4

21 synchronously integrated rates 8 held encoding rates

R1: E6 R2: E1 R3: E3 R4: E4

Every rate variable (29 rows; no row rescaling)

E1 E2 E3 E4 E5 E6 — Position E
D_CW D_CCW — Direction
S1 S2 — Step
J_CW1 J_CW2 J_CCW1 J_CCW2 — Joint
V_CW1 V_CW2 V_CCW1 V_CCW2 — Readout V
G_CW1 G_CW2 G_CCW1 G_CCW2 — Gate G
M1 M2 M3 M4 M5 M6 — Shared M
I_global — Global I

0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=6, D=-1, S=1 | E6 -> E5 -> E4 -> E3

21 synchronously integrated rates 8 held encoding rates

R1: E6 R2: E5 R3: E4 R4: E3

E1
E2
E3
E4
E5
E6
D_CW
D_CCW
S1
S2
J_CW1
J_CW2
J_CCW1
J_CCW2
V_CW1
V_CW2
V_CCW1
V_CCW2
G_CW1
G_CW2
G_CCW1
G_CCW2
M1
M2
M3
M4
M5
M6
I_global

Position E
Direction
Step
Joint
Readout V
Gate G
Shared M
Global I

Every rate variable (29 rows; no row rescaling)

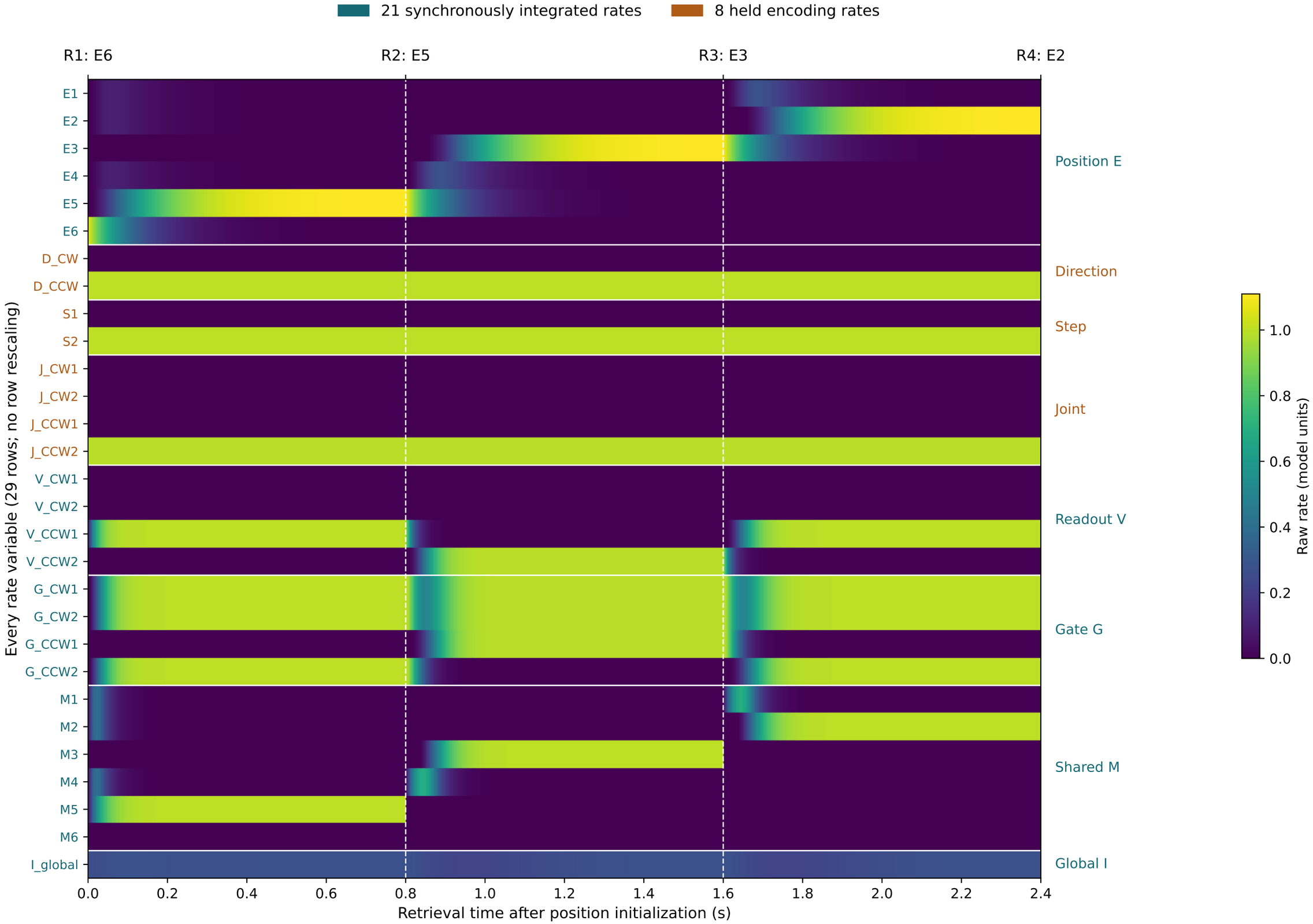
0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Retrieval time after position initialization (s)

Raw rate (model units)
0.0 0.2 0.4 0.6 0.8 1.0

V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.

All 29 rate variables | L=6, D=-1, S=2 | E6 -> E5 -> E3 -> E2



V/G/E/I carry across rank boundaries; shared M resets. All conditions use this same raw-rate color scale.
D/S/J are actual held rates. M is an effective shared drive; branch rectifications are not extra plotted neurons.